

Image-Similarity-Based Scale Estimation for 3D Model

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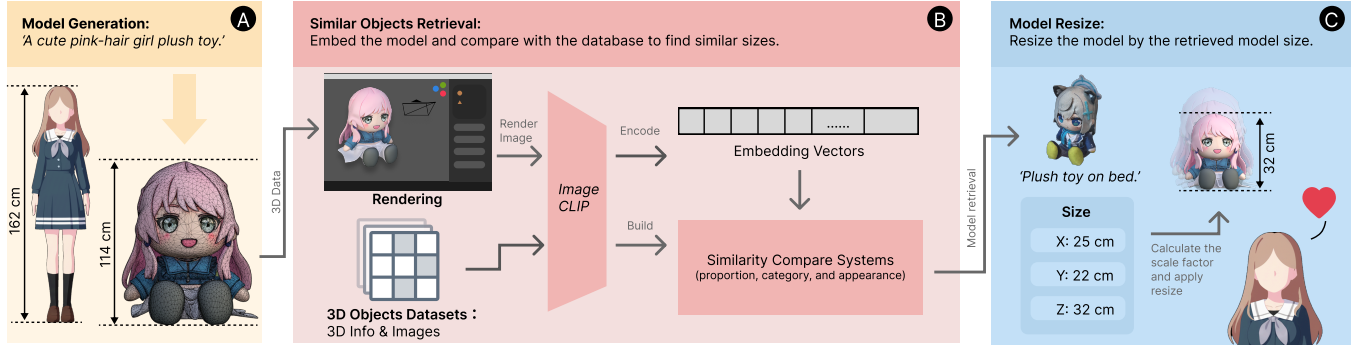


Figure 1: The workflow of our resizing approach is as follows: (A) Extract 3D information from the source model; (B) Render the object into an image, generate its embedding, and retrieve similar objects from the database; (C) Use the retrieved objects to resize the original model.

Abstract

Accurate object scaling is essential in building 3D scenes, yet manual adjustment is time-consuming for developers and hard to calculate, and 3D models imported from various sources often lack consistent scaling. To address this challenge, we propose an image-similarity-based scale estimation method that can automatically predict appropriate dimensions for 3D models using geometric and visual embeddings. The method shows superior accuracy and speed compared to baseline depth-based approaches. It offers an efficient solution for automated scene generation and model normalization.

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1 Introduction

With the application of generative models in 3D asset creation [Öcal et al. 2024; Stacchio et al. 2025] and accessibility of 3D asset libraries, users can rapidly generate wanted scenes and objects. However, models from online libraries, marketplaces, or AI-generated content frequently lack matched scale, as assets' sizes are sometimes normalized during production or generation. This requires time-consuming manual adjustments. Existing workflows typically involve comparing target objects to a human mannequin — a process that is both inefficient and imprecise.

To address this gap, we explored two straightforward approaches first. One was to directly prompt an LLM to guess object dimensions, but the results were unsatisfactory. The other was a depth-estimation pipeline: because 3D scene generation produces effect images, we applied depth estimation models to those renders and—given known camera intrinsics and extrinsics—converted depth estimates into physical object dimensions for scaling. Due to its limited performance and lack of semantic information, we propose an alternative approach. We propose an image-similarity-based 3D object scale estimation method. Given a normalized 3D object, our method will retrieve visually and geometrically similar models from a pre-built embedding database (using CLIP features) and estimate the appropriate scale based on their scenarios.

We evaluate our method through comparative experiments against a depth-based baseline, demonstrating superior accuracy and applicability for virtual scene synthesis.

2 Our Approach

The base idea is that the system will encode the 3D objects' appearance, use this information to match similar objects in the database, and finally use the empirical information in the database to adjust the size of objects.

Rendering Front-view Image. To get a general view of a 3D object, the system will render an image of the object with a set of parameters. In this scenario, we decide the view angle is frontal, slightly tilted upward, and to the left. By this operation, the system can get more three-dimensional information from the object. The workflow and the effects image are shown in Fig. 1.

Computing Embeddings and Database Building. The core objective of this step is to vectorize the rendered images, converting them into high-dimensional numerical vectors for subsequent similarity computation and data retrieval. This study adopts the CLIP (Contrastive Language-Image Pre-Training) model, which generalizability and robustness have been demonstrated across a wide range of applications.

Using this approach, to ensure consistency and comparability in subsequent similarity retrieval tasks, this study utilizes the 3D-FUTURE (3D Furniture shape with TextURE) dataset [Fu et al. 2021], a large-scale collection of realistic indoor furniture models with accurate geometric and texture information. Following the two aforementioned steps, a database comprising 16,565 models was constructed, with the data structured as shown in Fig. 2.

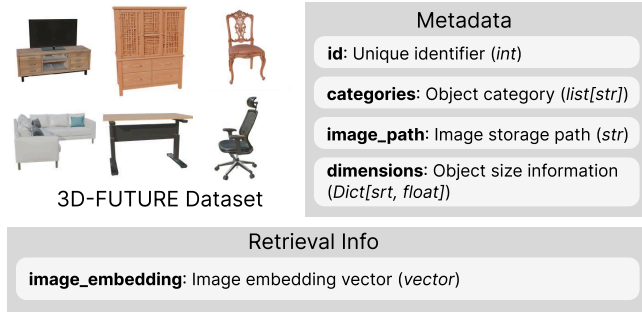


Figure 2: The structure of database, the 3D data and meta data is from 3D-FUTURE database [Fu et al. 2021].

Performing Similarity Retrieval. Given the embedding of the target object's rendered image, we perform similarity retrieval in a pre-built database storing embeddings, scales, and actual dimensions. The process is as follows:

- (1) Retrieve the top K most similar objects via cosine similarity of embeddings.
- (2) Compute cosine similarity between their dimensions and the target object.
- (3) Use exponentiated similarity weights to calculate the average estimated dimension.

Object dimensions (width, depth, height) are treated as 3D vectors, making cosine similarity effective for proportional comparison. Exponentiation is used instead of direct averaging to give greater influence to objects with more similar proportions.

Table 1: Evaluation Results

Method	Average Error	Variance	Confidence Interval (95%)	Usability
Depth	0.894	4.252	1.462 ± 0.810	5%
Ours	0.202	0.099	0.202 ± 0.098	80%

Scaling Strategy. The object is scaled based on the estimated average dimensions. Directly matching all three axes may lead to shape distortion. To avoid this, we use the shortest axis of the original object as the reference, as it often reflects key functional features.

Let object A be the target to scale and object B be the reference with average dimensions. Denote their dimensions as $A(l_a, w_a, h_a)$ and $B(l_b, w_b, h_b)$. The scaling factor:

$$s = \frac{\arg \min_{axis \in \{l, w, h\}} (axis_b)}{\arg \min_{axis \in \{l, w, h\}} (axis_a)} \quad (1)$$

Then, apply uniform scaling to all axes:

$$A'(l'_a, w'_a, h'_a) = (s \cdot l_a, s \cdot w_a, s \cdot h_a) \quad (2)$$

3 Evaluation

To evaluate the method, we constructed a test set of 40 normalized models across 8 categories (5 styles each), with ground-truth sizes manually obtained as a baseline for denormalization evaluation.

Since absolute dimensional errors are not comparable across objects of varying sizes, we adopt relative error—defined as the ratio of error to target dimension—as the evaluation metric. Given human perceptual tolerance for size variation at certain distances, errors below 15% are considered usable.

The results are summarized in Table 1. The depth estimation method yielded a high average error in validation. The low accuracy comes from error accumulation across multiple estimation steps.

In contrast, the CLIP-based denormalization method achieved a high usability of 80%. The negative cases involved atypical designs (e.g., an open-drawer cabinet). Overall, the CLIP-based method demonstrates significantly better accuracy and robustness across common objects, with potential for further improvement through dataset expansion.

4 Conclusions and Future Works

The similarity retrieval-based denormalization method shows good usability and visual consistency in most cases, but its performance depends on the size and diversity of the reference database. To enhance scalability and efficiency, future work could use neural networks to directly predict object dimensions from images, reducing reliance on retrieval and storage. This method is integrated into our agent-based 3D generation research, and we will evaluate its usability and efficiency (e.g., via manipulation-step comparisons) in the future.

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